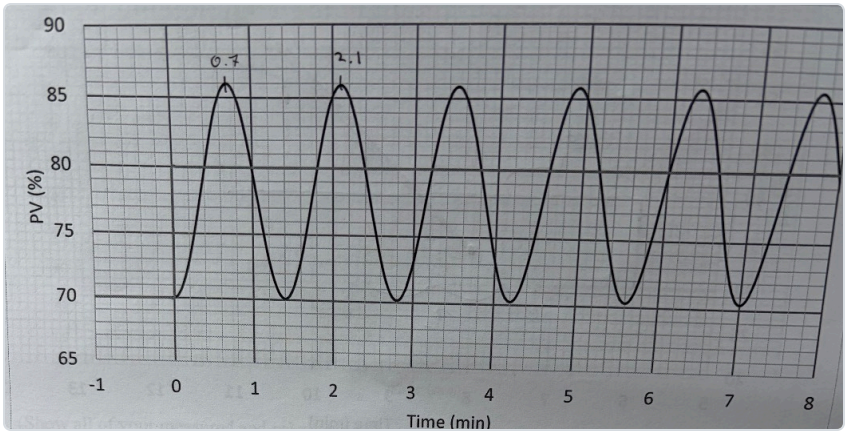
Controller Tuning Worksheet: Answer Key

CNTR 2371 · ZN closed loop, lambda, ZN open loop and decay ratio. Values were read from photos of the charts, so your own readings may differ by a few tenths; use the same steps with your numbers.

Q1 · Ziegler–Nichols closed loop (P-only, Kc = 1.55, SP +10%)



a) Measured values

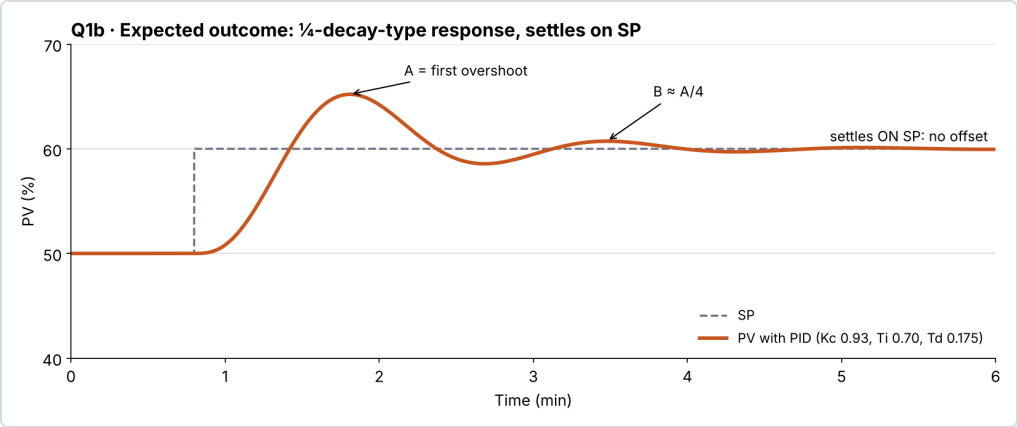
Reading	Value	Why
Peak / trough heights	≈ 86.5% / ≈ 70%, the same every cycle	sustained oscillation, DR = 1
Kcu	**1.55**	the gain that gave sustained oscillation
To (peak to peak)	2.1 − 0.7 = **1.4 min**	period of the oscillation

Calculated (PID row)

```
Kc = 0.6 × Kcu = 0.6 × 1.55 = 0.93
Ti = To / 2    = 1.4 / 2     = 0.70 min/repeat
Td = To / 8    = 1.4 / 8     = 0.175 min   (≥ 0.1 min, so PID, not PI)
```

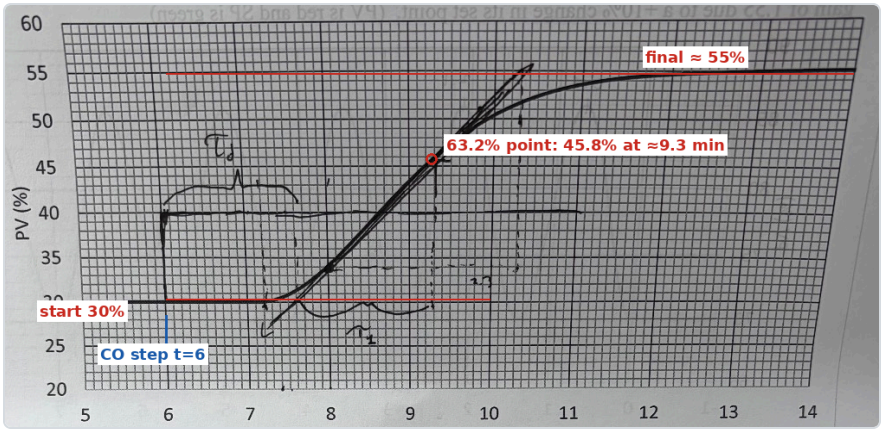
> **Answer:** PID with Kc = 0.93, Ti = 0.70 min/repeat, Td = 0.175 min.

b) Expected outcome (sketch)



Label: **A** (first overshoot), **B ≈ A/4** (second overshoot), and "settles on SP, no offset (integral action)".

Q2 · Lambda tuning (+10% CO at t = 6 min)



Measured values

Reading	Value
Start / final PV	30% → ≈ 55%, so ΔPV = 25%
Dead time τd	**1.5 min** (from the CO step at 6 min to where the PV starts to rise, about 7.5 min)
63.2% point	30 + 0.632 × 25 = 45.8%, reached at ≈ 9.3 min
Time constant τ1	9.3 − 7.5 = **1.8 min**
Process gain Kp	ΔPV / ΔCO = 25 / 10 = **2.5**

a) Transfer function and aggressive control (λ = 1)

```
Gp(s) = Kp·e^(−τd·s) / (τ1·s + 1) = 2.5 e^(−1.5s) / (1.8s + 1)

Larger of τ1 (1.8) and τd (1.5) = 1.8   →   τc = 1 × 1.8 = 1.8
Kc = τ1 / [Kp(τc + τd)] = 1.8 / [2.5(1.8 + 1.5)] = 1.8 / 8.25 = 0.22
Ti = τ1 = 1.8 min/repeat
```

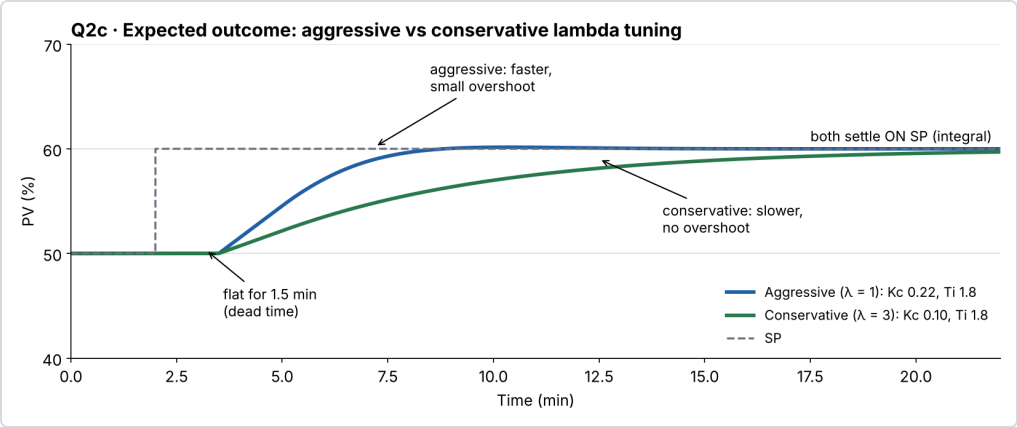
b) Conservative control (λ = 3)

```
τc = 3 × 1.8 = 5.4
Kc = 1.8 / [2.5(5.4 + 1.5)] = 1.8 / 17.25 = 0.10
Ti = τ1 = 1.8 min/repeat
```

Answer: Gp(s) = 2.5e^(−1.5s)/(1.8s + 1). Aggressive: Kc = 0.22, Ti = 1.8 min. Conservative: Kc = 0.10, Ti = 1.8 min.

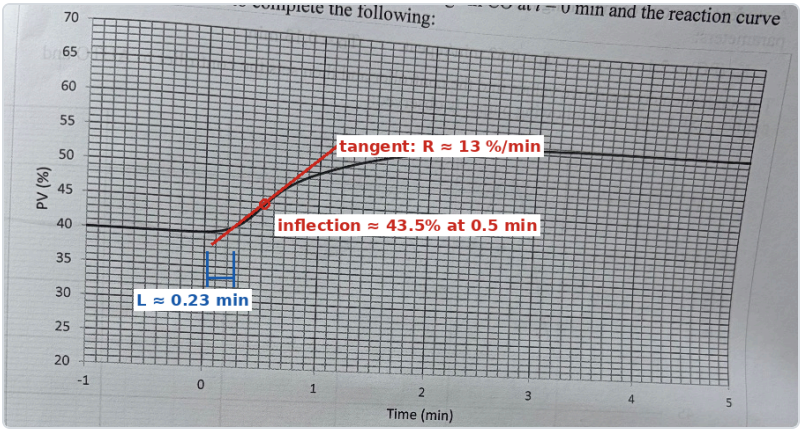
The reaction rate R (≈ 9.1 %/min) belongs to the ZN open-loop method. Lambda tuning does not use it.

c) Expected outcome (sketch)



Label both curves. Both stay flat for the 1.5 min dead time. Aggressive rises faster (a small overshoot is possible); conservative is slower with no overshoot. Both settle on SP.

Q3 · ZN open loop / reaction curve (+10% CO at t = 0)



a) Reaction rate and dead time

Reading	Value
Starting PV	40%
Inflection (steepest point)	≈ 43.5% at ≈ 0.5 min
Reaction rate R (tangent slope)	**≈ 13 %/min**
Dead time L (tangent crosses 40%)	**≈ 0.23 min**
LR = R × L	13 × 0.23 ≈ **3.0%**

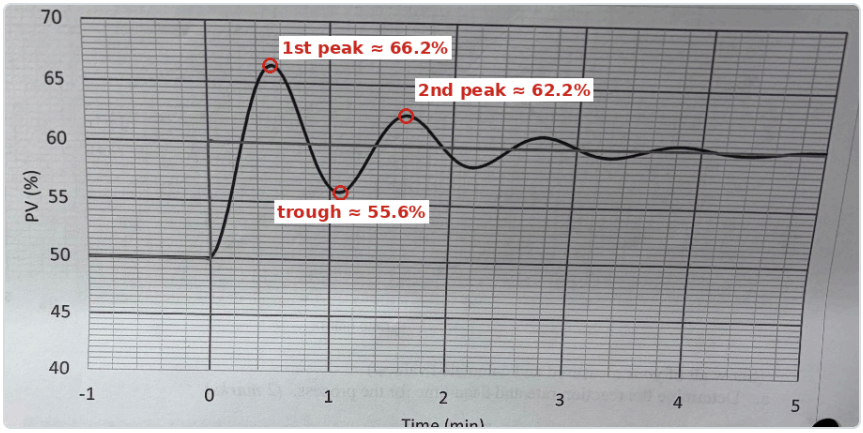
b) Tuning parameters

```
Kc = 1.2 × ΔCO / LR = 1.2 × 10 / 3.0 = 4.0
Ti = 2L   = 2 × 0.23   = 0.46 min/repeat
Td = 0.5L = 0.5 × 0.23 = 0.115 min   (≥ 0.1 min, so PID is allowed)
```

Answer: PID with Kc = 4.0, Ti = 0.46 min/repeat, Td = 0.115 min.

Td is right at the 0.1 min limit. If your L reads under 0.2 min, Td comes out under 0.1 min, so use PI instead: Kc = 0.9 × ΔCO / LR (3.0 with LR = 3.0) and Ti = 3.33 L.

Q4 · Checking a student's tuning (Kc 2.1, Ti 0.52, Td 0.13)



a) Decay ratio

```
A = 1st peak − trough       = 66.2 − 55.6 = 10.6
B = 2nd peak − same trough  = 62.2 − 55.6 = 6.6
DR = (B / A)² = (6.6 / 10.6)² = 0.39

Check, measured from the final value (≈ 60%):
DR = (62.2 − 60) / (66.2 − 60) = 2.2 / 6.2 = 0.35
```

Answer: DR ≈ 0.39, which is more than ¼, so the response is too oscillatory.

b) Kc

Decrease Kc. Less proportional action gives a more stable loop, so the oscillations die out faster.

c) Keeping Kc = 2.1

Increase Ti (slower integral, so it adds less lag and the loop is more stable). **Increase Td by the same factor** to keep Rate/Reset = ¼ (now 0.13 / 0.52 = ¼, so keep Td = Ti/4).